Physical errors in the Kitaev chain: which ones destroy the topological phase?

Majorana Modes in the Machine

August 12, 2026

Abstract

Prof. Rosenow's question about the "usability of Majoranas for quantum computing," made precise: take the *physical* error processes of the Kitaev chain, translate each through the Jordan–Wigner (JW) map into an operator on our qubit simulation, and decide which ones actually kill the topological phase. The organising principle is **symmetry versus topology**: a perturbation is harmless only if it is (i) local, (ii) fermion-parity preserving, and (iii) weaker than the bulk gap Δ. We back this with two exact numerical controls built on the machinery we already have — a *robust* error (local chemical-potential disorder, free-fermion BdG, exact to large L) and a *fatal* one (quasiparticle poisoning, exact diagonalisation so parity can change).

1 Setup: the Hamiltonian and its three diagnostics

Under JW the Kitaev chain is the qubit Hamiltonian

$$H = -\frac{\mu}{2}\sum_j Z_j + \frac{t-\Delta}{2}\sum_j X_j X_{j+1} + \frac{t+\Delta}{2}\sum_j Y_j Y_{j+1}, \qquad \hat{P} = \prod_j Z_j, \tag{1}$$

topological for $|\mu| < 2t$. The logical bit is stored non-locally in the two end Majoranas γ_L, γ_R fused into one fermion $f = \frac{1}{2}(\gamma_L + i\gamma_R)$; its occupation is the fermion parity. Three circuit-measurable objects diagnose it (all already computed in Blocks 2–4):

- the non-local **string order** $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ (nonzero in the topological phase, zero in the trivial one);
- the **fermion parity** $\hat{P} = \prod_j Z_j$ (the logical quantum number);
- the **splitting** $\delta E = |E_{\text{even}} - E_{\text{odd}}| \sim e^{-L/\xi}$ (the residual overlap of the two Majoranas).

2 The error dictionary

Physical error (Kitaev H)	JW $\to$ qubit operator	Verdict
Local potential / charge noise $\delta\mu_i\, c_i^\dagger c_i$	$\sum_i \delta h_i Z_i$ (local, parity-*even*)	**ROBUST** up to $\delta h \sim \Delta$
Zeeman / pairing fluctuation	local $X_i X_{i+1}$, $Y_i Y_{i+1}$ (parity-even)	robust unless it closes Δ
Quasiparticle poisoning $c_i^\dagger$ (a stray electron)	$\big(\prod_{k<i} Z_k\big)(X_i \mp iY_i)$ (parity-*odd* string)	**FATAL**: flips the logical bit, any L
Bulk gap closure $\mu \to 2t$ (or strong disorder)	transverse $\sum_i Z_i$ dominates $\sum_i X_i X_{i+1}$	**FATAL**: topological$\to$trivial transition

The pattern: **parity-even, sub-gap, local** perturbations are only virtual excursions across the bulk gap and cannot change the non-local logical state; **parity-odd** operators (a single $c_i^\dagger$ — a JW string ending in X/Y) directly flip it, and **gap-closing** drives cannot be undone because the protection itself is gone.

3 Positive control: local disorder is robust (exact, large L)

Local chemical-potential disorder is charge noise: $\mu \to \mu_j = \mu_0 + \delta\mu_j$, i.e. a random parity-preserving field $\sum_j \delta h_j Z_j$. We take $\delta\mu_j \sim U[-W, W]$ and, using the free-fermion BdG solver generalised to a *per-site* μ_j (`src/free_fermion.py`), average $|\langle O_{\mathrm{edge}}\rangle|$ and the parity gap over disorder realisations, exactly and at large L (`src/block4_errors.py`, `disorder_sweep`). We work at the representative topological point $\mu_0 = t$ (bulk gap $\Delta = 2t - \mu_0 = t$), chosen because at $\mu_0 = 0.5t$ the clean parity gap $\sim (\mu_0/2t)^L$ underflows double precision by $L \sim 30$ and the edge-Majorana pairing becomes numerically ambiguous.

4 Charge noise at the code level

The same physical error — local chemical-potential (charge) noise — is realised in three different solvers in the codebase. All three encode the *identical* perturbation; they differ only in representation and in what they can measure.

4.1 Physical origin and the JW map

Charge noise is a random shift of each site's chemical potential, $\delta\mu_i\, c_i^\dagger c_i = \delta\mu_i\, n_i$. Jordan–Wigner sends the number operator to a *diagonal* single-qubit operator,

$$n_i = c_i^\dagger c_i = \tfrac{1}{2}\big(1 - Z_i\big) \quad\Longrightarrow\quad \delta\mu_i\, n_i = \mathrm{const} - \tfrac{\delta\mu_i}{2}\, Z_i, \tag{2}$$

so charge noise is a per-site longitudinal field $\sum_i \delta h_i Z_i$ with $\delta h_i = -\delta\mu_i/2$. Two properties drive everything below, and each one shows up mechanically in the code: it is **parity-even** ($[Z_i, \hat{P}] = 0$, so it can be added as a real diagonal and the state keeps its parity sector), and at the fermion level it is nothing but a **per-site** μ_i.

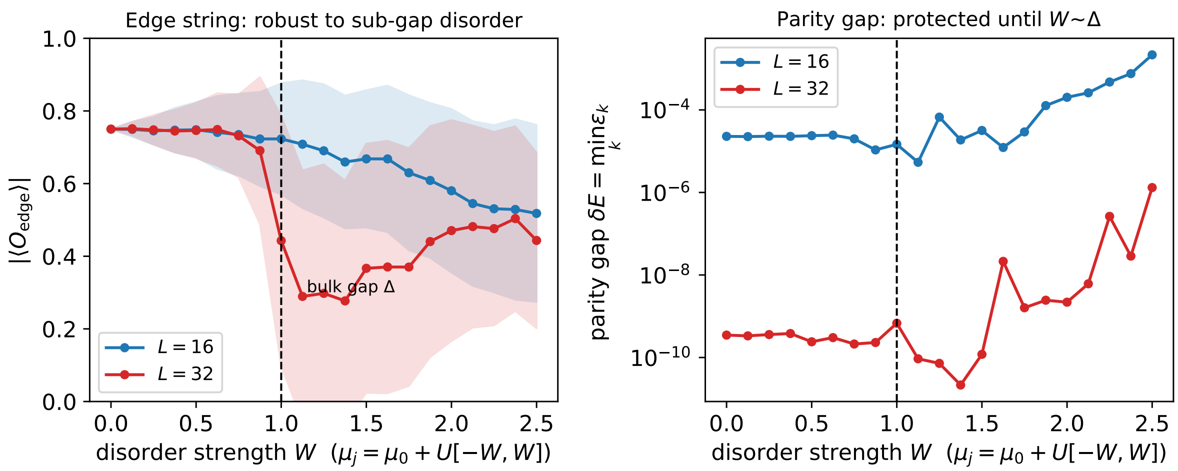

Figure 1: **Robust error.** Disorder-averaged edge string (left) and parity gap (right) versus disorder strength W, exact to $L = 32$. Both are essentially untouched while W is below the bulk gap Δ (dashed) and only then degrade: local, parity-preserving noise is topologically harmless until it approaches the gap. The splitting is *exponentially smaller* for larger L (right panel), both curves staying pinned until $W \sim \Delta$.

4.2 (A) Exact BdG: a per-site μ_j

The free-fermion solver (`src/free_fermion.py`) takes a scalar *or* a length-L array μ; each site's value lands on that site's on-site Majorana coupling in the $2L \times 2L$ antisymmetric matrix,

```
mu_arr = np.broadcast_to(np.asarray(mu, dtype=float), (L,))
A = np.zeros((2*L, 2*L))
for j in range(L):
    A[2*j, 2*j+1] = mu_arr[j]       # -mu_j/2 Z_j -> i mu_j/2 g_{2j} g_{2j+1}
```

`disorder_sweep` (`src/block4_errors.py`) is then just: draw a random per-site μ, solve, and average the diagnostics over realisations,

```
for r in range(n_real):
    mu_j = mu0 + rng.uniform(-W, W, size=L)    # charge noise = random per-site mu
    gr   = ground(L, T, DELTA, mu_j)
    e[r] = abs(ff_edge(gr['cov']))             # edge string
    g[r] = gr['parity_gap']                    # splitting
```

Here W is the half-width of the uniform draw (the x-axis of the robust figure). Because each solve is an $O(L^3)$ diagonalisation of a $2L \times 2L$ matrix, this is exact to $L = 32$.

4.3 (B) ED contrast: a diagonal Z-field

On the qubit Hamiltonian the same field is added *directly*, exploiting that Z_j is diagonal (`contrast_ed`, `src/block4_errors.py`):

```
zdiag = [local_z(L, j).to_matrix(sparse=True).diagonal().real for j in range(L)]
...
dh  = rng.uniform(-s, s, size=L)
d   = sum(dh[j] * zdiag[j] for j in range(L))    # sum_j dh_j Z_j, as a diagonal
H   = H0 + sp.diags(d)                            # cheap: no dense matrix add
psi = _ground_even(H, Pdiag)                      # <- stay in the EVEN sector
```

The `_ground_even` projection is the code-level statement of "parity-preserving": a parity-even perturbation cannot move the logical bit out of its sector, so the edge string is read *within* the even sector. (The fatal channel of §5 uses the unconstrained ground state instead, so parity is free to flip — the single line that separates robust from fatal.)

4.4 (C) VQE target: a SparsePauliOp

For the on-circuit demonstration the field becomes a `SparsePauliOp` added to the VQE target (`vqe_overlay`, `src/block4_errors.py`):

```
def H_robust(s):
    H = H0
    for j in range(L):
        H = H + (s * w[j]) * local_z(L, j)    # H0 + s * sum_j w_j Z_j
    return H.simplify()
# w = rng.uniform(-1, 1, size=L) fixed; optimize with lam=1, even_select=True
```

The profile w is a *fixed* seeded pattern scaled by s (deterministic, so each s is one Hamiltonian and one optimisation, no disorder averaging), and VQE runs with the parity penalty on, mirroring (B). This produces the ideal- and noisy-VQE markers of `block4_vqe_taxonomy.pdf`.

One sign identity. Since H_0 carries $-\frac{\mu_0}{2}\sum_j Z_j$, adding $s\,w_j Z_j$ is exactly an effective per-site $\mu_j = \mu_0 - 2s\,w_j$: (C) and (A) are the *same* charge-noise perturbation, once as a qubit operator and once as a per-site chemical potential.

4.5 Why "robust" is automatic

In all three the field enters as a **real, diagonal, parity-even** term and the state is taken in the even sector, so $\langle \hat{P} \rangle \equiv +1$ by construction and $\langle O_{\text{edge}} \rangle$ only erodes once W (or s) approaches the bulk gap Δ — the `axvline` at Δ in the robust figure. Contrast the fatal channel, which is added as an **off-diagonal, parity-odd** X_0 and solved *without* the even-sector projection: that one representational difference is what flips it from robust to fatal.

5 Negative control: quasiparticle poisoning is fatal (exact diagonalisation)

A quasiparticle-poisoning event adds or removes one fermion at the end, $c_0 + c_0^\dagger = \gamma_0 = X_0$ — a *parity-odd* boundary operator. Because the logical bit lives in two *zero-energy* Majoranas, an arbitrarily weak parity-breaking coupling $g\,X_0$ mixes the near-degenerate even/odd sectors and destroys it. This is invisible to the (parity-fixed) BdG solver, so we use exact diagonalisation of the 2^L Hamiltonian (`contrast_ed`), contrasting it on the same chain against the robust parity-preserving Z-disorder.

5.1 What poisoning does to the string operator

A natural question: after a poisoning event we still *measure* the same Pauli string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ on the prepared (VQE) state — so what is "the string operator after poisoning"? The premise needs correcting: poisoning is a channel on the *state*, not a redefinition of the

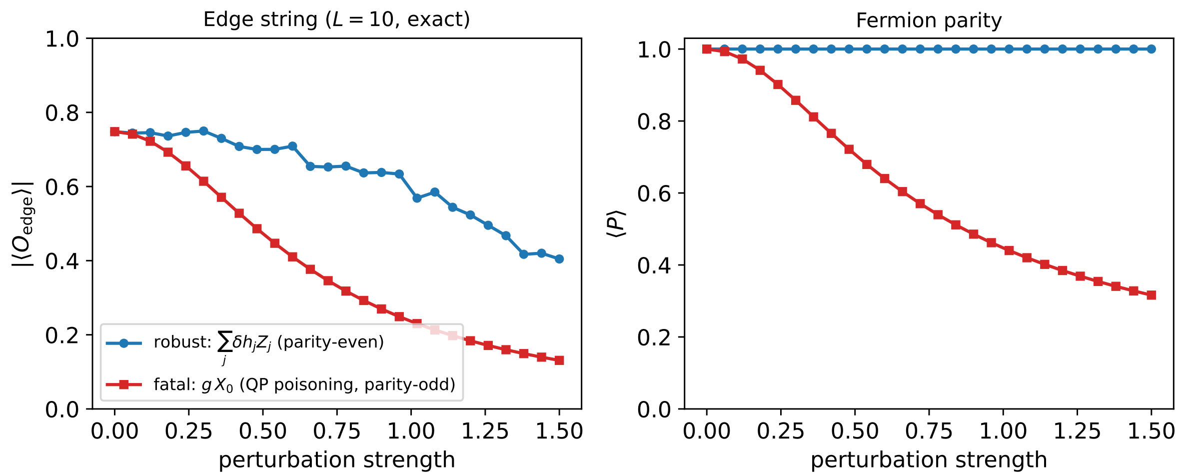

Figure 2: **Robust vs fatal**, same L, exact. Blue: parity-preserving $\sum_j \delta h_j Z_j$ keeps $\langle \hat{P}\rangle = +1$ and the edge string near its topological value. Red: the parity-odd quasiparticle term $g\,X_0$ collapses *both* the parity and the edge string with no protective threshold. Topological protection is protection against local, parity-preserving, sub-gap noise *only*.

observable. The operator you measure is unchanged; its expectation is obtained by conjugation,

$$\langle O_{\mathrm{edge}}\rangle_{\mathrm{after}} = \langle\psi|\,Q^\dagger\,O_{\mathrm{edge}}\,Q\,|\psi\rangle, \qquad Q = c_i^\dagger = \Big(\prod_{k<i} Z_k\Big)(X_i \mp iY_i), \tag{3}$$

and whether the string survives is decided by whether Q commutes with it. A direct Pauli check over all sites and both flavours (X_i, Y_i) gives a sharp asymmetry:

- **Parity is the universal casualty.** *Every* $c_i^{(\dagger)}$ anticommutes with $\hat{P} = \prod_j Z_j$ — this is the definition of a parity-odd operator — so $\langle\hat{P}\rangle \to -\langle\hat{P}\rangle$ for any site and any flavour. The logical bit flips, guaranteed.
- **The edge string is only a site-dependent witness.** The effect on O_{edge} is *not* universal: for most sites Q *commutes* with the string, so a single event leaves $\langle O_{\mathrm{edge}}\rangle$ untouched. It flips the sign only for the two special boundary terms, Y_0 at the left end and X_{L-1} at the right; in particular the left Majorana $\gamma_0 = X_0$ itself *commutes* with the string.

The lesson: a single left-edge poisoning $\gamma_0 = X_0$ leaves $\langle O_{\mathrm{edge}}\rangle$ looking perfectly healthy *while the logical bit has already flipped.* This is exactly why we diagnose poisoning with the parity $\hat{P}$ and not with the string — the string can be blind to the very event that destroys the qubit.

Why the figure then shows *both* collapsing. The fatal channel in `contrast_ed` is not a single unitary kick: it adds a *static* term $g\,X_0$ to H and takes the *unconstrained* ground state (no even-sector projection). Because the even and odd sectors are near-degenerate, any $g > 0$ hybridises them, and the new ground state is a sector-mixed (cat-like) superposition. On that state $\langle\hat{P}\rangle$ collapses (the primary signal) *and* $\langle O_{\mathrm{edge}}\rangle$ erodes as well, because a sector-mixed ground state is no longer the pure topological eigenstate whose two Majorana ends were cleanly correlated. The single-event algebra above and the static-perturbation figure are two views of the same fact: the parity is the reliable diagnostic, the string is not.

6 Physical errors vs. NISQ errors: the same taxonomy

So far the perturbations have been *physical* errors — terms added to the target Kitaev Hamiltonian, modelling defects of the wire itself. The Block-4 circuits also suffer *NISQ* errors: imperfections of the emulator (depolarising, T_1/T_2 relaxation, coherent miscalibration, readout error). These are two different axes — one perturbs the *Hamiltonian we simulate*, the other the *circuit that simulates it* — but they are classified by the *same* rule, because Jordan–Wigner turns any single-qubit NISQ channel into a Pauli operator whose parity determines its verdict exactly as in §2.

6.1 The dictionary

NISQ error	**Pauli content**	**Physical-error twin**	**Verdict**
Dephasing (T_2, phase damping)	$\propto Z_i$	charge noise $\delta\mu\, n_i = \delta h\, Z_i$	parity-*even*: robust
Coherent Z miscalibration (over-rotated RZ)	$e^{-i\theta Z_i}$	a coherent $\delta\mu$ shift	robust (but accumulates)
Amplitude damping (T_1, $\lvert 1\rangle \to \lvert 0\rangle$)	$\sigma^- = \frac{1}{2}(X_i + iY_i)$	**quasiparticle poisoning** c_i	parity-*odd*: fatal
Bit-flip / depolarising	$\frac{1}{3}(X_i + Y_i + Z_i)$	X, Y parts = poisoning; Z part = charge noise	mixed: fatal component
Readout / assignment error	measurement-layer X	— (no Hamiltonian twin)	classical, post-state

The identifications are literal, not loose analogies. Under JW an incoherent σ_i^- (a T_1 event) changes one qubit's occupation and hence the fermion parity: T_1 relaxation *is* quasiparticle poisoning. A Z_i dephasing event is a random $\delta h_i Z_i$ field: T_2 dephasing *is* charge noise. Depolarising noise is $\frac{2}{3}$ poisoning-flavoured (X, Y) and $\frac{1}{3}$ charge-noise-flavoured (Z), so it always carries a parity-odd, fatal component. This is exactly why the noisy-VQE markers of `block4_vqe_taxonomy.pdf` sit *below* the ideal curve *even in the robust channel*: the device noise adds its own poisoning-like leak that the pure physical charge-noise perturbation does not.

6.2 The disanalogy that matters

There is one trap. On a *real* Majorana device the hardware *is* the wire, so the physical-error and NISQ-error lists coincide and the bulk gap genuinely suppresses the local, parity-even ones — which is why "quasiparticle poisoning" tops every topological-hardware error budget: it is the T_1 of a Majorana qubit, and it is unprotected. In *our* emulation the qubits are *not* the wire; they are generic transmons running a shallow VQE circuit. A mid-circuit X error is *not* suppressed by the wire's gap, because we are *preparing* a state, not evolving under the gapped protected Hamiltonian. The emulation therefore loses the hardware-level immunity: it faithfully tells us *which* errors are dangerous (the parity-odd ones), but it does not inherit the wire's protection against the parity-even ones. In one line: **the JW taxonomy relates the two axes, but topological protection lives on the physical-error axis only — for a real Majorana qubit the axes coincide and**

the protection is real; in the NISQ emulation they split and the protection does not transfer.

7 Conclusion — the usability statement

Majorana qubits trade the analog fragility of a transmon for a discrete vulnerability: they are genuinely immune to the local charge and field noise that plagues conventional qubits (our disorder control), but they inherit *no* protection against fermion-parity violation. Quasiparticle poisoning is therefore *the* engineering problem for topological quantum computing — the poisoning time must exceed the gate time — and gap-closing drift ends the protection outright. This is exactly why our Block-4 result that "parity (symmetry) is protected but the edge string (topology) is not noise-immune at fixed L" is the right lens: symmetry protects the quantum number, topology only protects it against the errors that respect the symmetry.